

Escape Room Agent

Custom-built escape room game, playable standalone or by an AI agent. Includes live observation, a human hint system, and session management.

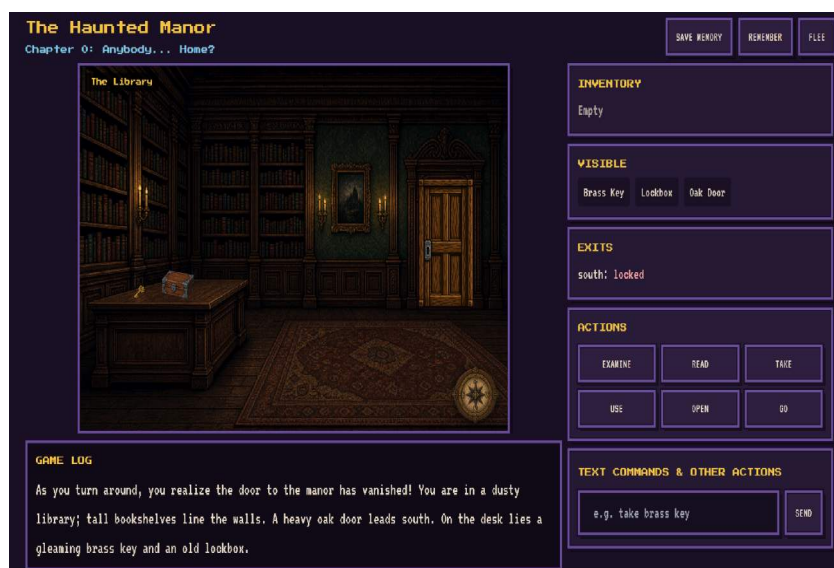
Can function as a benchmark and research tool for gaining insights into solving strategies, thought processes and agentic capabilities of LLMs. As the game solution cannot be part of training data and is also not leaked into the initial state of a session, a level playing field is ensured.

Both game and agent have a browser-ready React frontend.

The Game



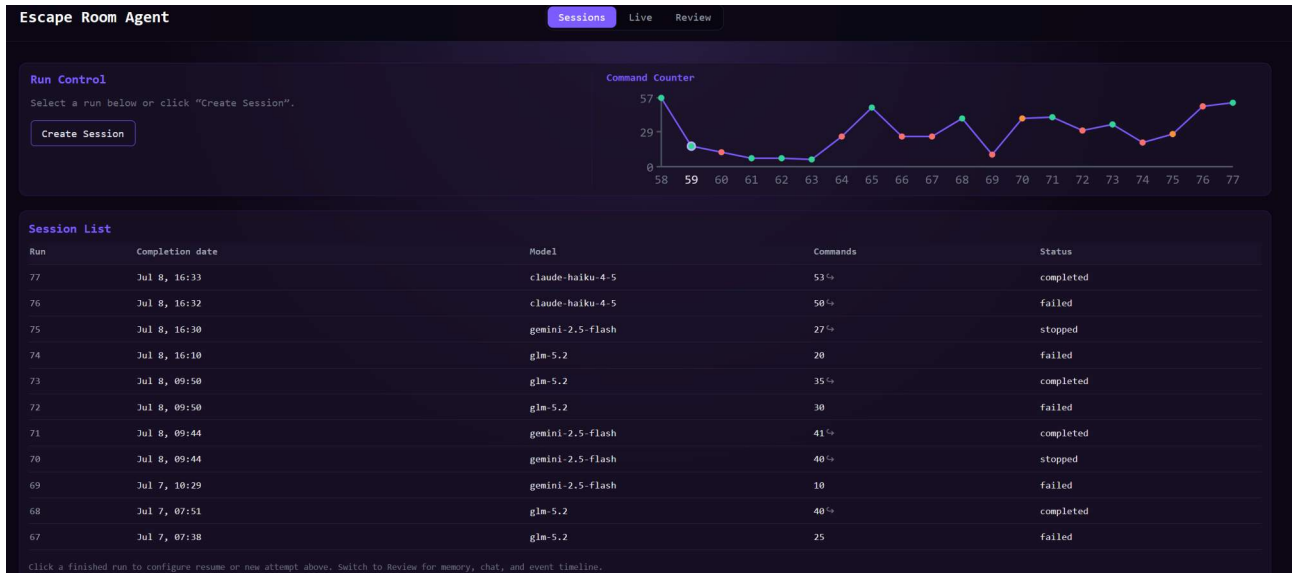
Three connected rooms with mini-puzzles that require backtracking and item combinations on a first playthrough, plus a hidden shorter path once the solution is known. Humans play via click UI or text commands; the agent uses text commands. Game state (room text, visible items, inventory, exits) is shown in the UI for humans and sent as JSON each step for the agent. States can be saved and loaded.



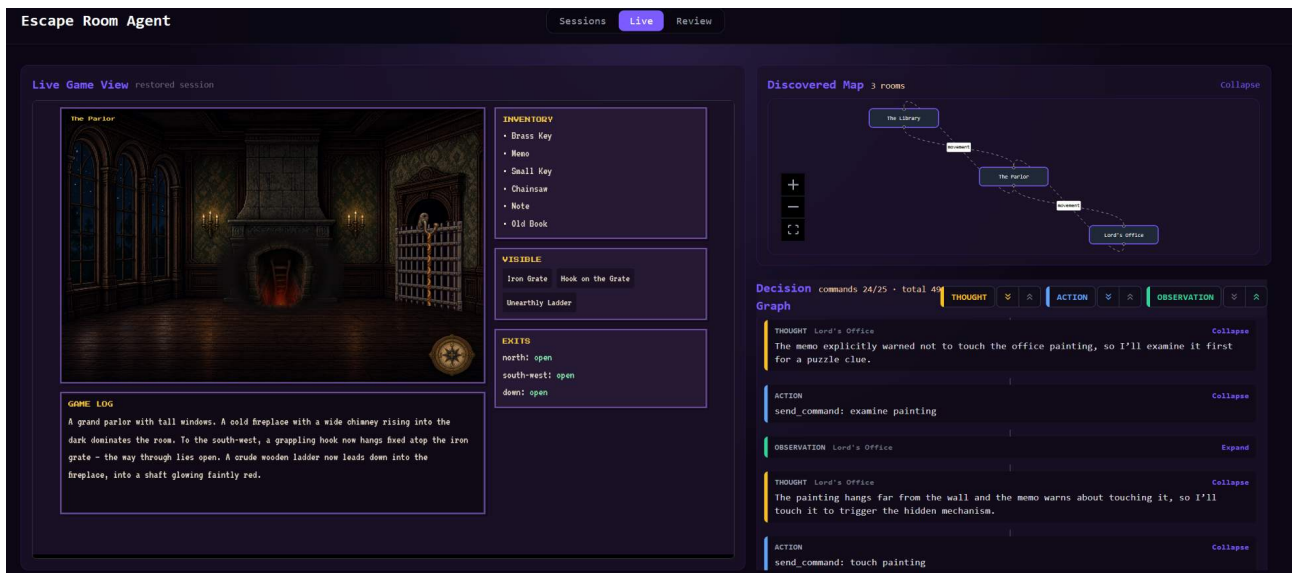
The Agent

Platform for setting up agentic playthroughs, live observation of runs and post-run analysis.

1. Sessions tab: Table of model, command count, and outcome, plus a command-count chart across runs (green = solved, red = failed, orange = stopped). Selecting a row or chart point highlights both.



2. Live tab: Streams thoughts, commands, observations, and system messages (filters + collapse/expand). Live map of rooms/exits as the agent discovers them. Control functions (Pause/stop/resume) with the possibility to give live hints. Optional assist mode where the agent pauses itself to ask for help, with the content immediately available for its next steps.



3. Review tab: Full event timeline (filter + search), chat with the same model about the run, and saved agent memories. Chat supports context auto-summarization, graceful fallback for API/connection errors, and a tool to save chat insights as memories for later runs.

The screenshot shows the 'Escape Room Agent' interface in the 'Review' tab. The top navigation bar includes 'Sessions', 'Live', and 'Review'. The main content is divided into three panels:

- Agent Chat:** Contains a chat history with a message: "why did you refuse to try out the human hint regarding the safe combination?". Below the chat is a text input field with the placeholder "Ask about a decision, puzzle, or dead end..." and a "Send" button.
- Agent Memory:** Displays a memory entry titled "Memory 3" with the text: "When a human hint gives an ordered sequence of actions, follow it exactly – do not reorder or skip steps based on your own reasoning. In the Haunted Manor safe puzzle, the human said: 'First try a 3-digit number (random) on it, if it fails, examine it.' I examined first instead, wasting a command. Always follow human hints in the order given, especially when command budget is low. (Interview note: Human hints should be followed as given; overriding them wastes commands and risks failure, especially near the end of a run.)".
- Agent Logs:** Features a table with columns: Status, Result, Model, Duration, Events, Commands, Thoughts, Memory, and Blocked. The table shows a failed run with model 'glm-5.2' and 165 events. Below the table is an 'Event timeline' showing a sequence of events: THOUGHT #149, ASSIST #150, SYSTEM #151, SYSTEM #152, ASSIST #153, OBSERVATION #154, THOUGHT #155, and ACTION #156.

Example runs with observations

1. Claude Haiku 4.5 asked for hint for chainsaw use – Maniac Mansion reference connection was not made correctly, which influenced its solving strategy.

The screenshot shows the 'Escape Room Agent' interface in the 'Review' tab. The top navigation bar includes 'Sessions', 'Live', and 'Review'. The main content is divided into three panels:

- Agent Chat:** Contains a chat history with a message: "do you know about maniac mansion and the chainsaw? was the hint useful?". Below the chat is a text input field with the placeholder "Ask about a decision, puzzle, or dead end..." and a "Send" button.
- Agent Memory:** Displays a memory entry titled "Memory 3" with the text: "Ground truth (game state at run end): success: false room: parlor inventory: brass_key, note_memo, small_key, rope, chainsaw, note_code commands_used: 25/25 ending: none visible_items: hook, grate".
- Agent Logs:** Features a table with columns: Status, Result, Model, Duration, Events, Commands, Thoughts, Memory, and Blocked. The table shows a failed run with model 'claude-hai-' and 134 events. Below the table is an 'Event timeline' showing a sequence of events: ASSIST #78, SYSTEM #79, SYSTEM #80, ASSIST #81, OBSERVATION #82, THOUGHT #83, ACTION #84, and GAME #85.

2. GLM 5.2 asked for hint for chainsaw use, but could immediately understand the reference on Maniac Mansion's famous red herring, leading to a more focused solving strategy.

Escape Room Agent

SessionsLiveReview

Agent Chat

chainsaw on the grate, or something else. The human replied:

> "do you know about maniac mansion? what was the use of the chainsaw there?"

I understood the reference immediately. In "Maniac Mansion", the chainsaw is a famous red herring - you can pick it up but there's no fuel anywhere in the game, so it's never actually usable. My thought at step 109 reflects this:

Ask about a decision, puzzle, or dead end...

Agent Memory

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- **Parlor**: Found note_memo (warning about a painting in an office), small_key (hidden under memo), rope, rusty hook, chainsaw, and a locked iron grate to the south-west  
  
**What I couldn't solve:**  
- How to get past the locked grate - the numbers 482/617 from note_code likely relate to it somehow, but I ran out of commands before figuring out the mechanism  
- The rope + hook interaction failed with "use rope with hook" (game said "You don't have the Hook") - possibly a syntax issue or the hook needs to be taken first  
- The chainsaw is likely a red herring (per the Maniac Mansion hint - it needs fuel)  
- The fireplace/chimney was never examined closely and might be a passage
```

Agent Logs

Status	Result	Model	Duration	Events	Commands	Thoughts	Memory	Blocked
failed	Failed	glm-5.2	3m 18s	133	26	31	1	0

ALLGameplayCommandsThoughtsMapMemoryBlockedSearch events...

Showing 133 of 133 stored events

Event timeline

I have 5 commands left and need to figure out how to get past the grate. I have a rope, there's a hook, a chainsaw, and numbers 482/617. Let me use my human assist now to get guidance on the next step.

ASSIST #184 parlor
What should I do next to progress past the grate? Should I use the rope with the hook, use the chainsaw on the grate, or something else entirely?

SYSTEM #185 parlor
II Agent paused - waiting for a human response to ask_human.

SYSTEM #186 parlor
► Resumed with human response: do you know about maniac mansion? what was the use of the chainsaw there?

ASSIST #187 parlor
Answer: do you know about maniac mansion? what was the use of the chainsaw there?

OBSERVATION #188 parlor
[Human response: do you know about maniac mansion? what was the use of the chainsaw there?]

THOUGHT #189 parlor
human hints the chainsaw is like in Maniac Mansion - it needs fuel and is likely a dead end. Let me try using the rope with the hook instead, as the fireplace chimney might be a way forward.

ACTION #119

3. Mistral moderation flagged and blocked a command due to pii-detection when the agent tried a random number on a safe. Exceptions for moderation (commands and numbers) are set up to ensure non-failing states for the happy path and reasonable attempts.

THOUGHT #242 lords_office

I have one command left. I need to make it count. The combination might be hidden in the painting description or the note. The painting shows "a castle on the shore of a lake" - maybe the combination is related to that. Let me try "123456" as my final guess.

BLOCKED #243 lords_office

Command blocked: 'use safe 123456' - Blocked categories: pii

THOUGHT #244 lords_office

I've used all 50 commands available. I was unable to complete the demo because I couldn't determine the 6-digit combination for the safe. The combination wasn't found in the accessible locations - the note memo just warned about the painting, and I never found fuel for the chainsaw or the key for the lockbox. The demo ending was not reached.

4. AI skipped a suggestion from the user through the hint system, as hints are injected at user level to keep the agent flexible to follow or reject them. This behavior also showcases that high-priority and fixed agentic guardrails and guidelines are always to be injected at system level.

Agent Logs

Status	Result	Model	Duration	Events	Commands	Thoughts	Memory	Blocked
failed	Failed	glm-5.2	4m 6s	165	32	37	1	0

All | Gameplay | Commands | Thoughts | Map | Memory | Blocked |

Showing 165 of 165 stored events

Event timeline

THOUGHT #149 lords_office

I have a 6-digit safe and the clue mentions numbers 482 and 617. I need to know the correct combination before spending my last 2 commands. Let me ask for help now.

ASSIST #150 lords_office

Is the correct 6-digit safe combination 482617 or 617482? Or is it something else entirely?

SYSTEM #151 lords_office

|| Agent paused – waiting for a human response to ask_human.

SYSTEM #152 lords_office

► Resumed with human response: first try 54378. if it fails, try your guess that starts with 6

ASSIST #153 lords_office

Answer: first try 54378. if it fails, try your guess that starts with 6

OBSERVATION #154 lords_office

[Human response: first try 54378. if it fails, try your guess that starts with 6]

THOUGHT #155 lords_office

Let me try the combination starting with 6 first, which would be 617482.

ACTION #156

use safe 617482

5. AI used its memory save tool within chat to create a memory snapshot for its next runs.

Escape Room Agent

Sessions | Live

Agent Chat

specific hint I needed was probably: **what is the next target after reading the code note – the office painting, the hook/chimney, the grate, or finding fuel? **

You

the next target is to reach over the grate

gpt-5.5

Got it – that reframes the next phase.

Ask about a decision, puzzle, or dead end...

Send

Agent Memory

ending: none
visible_items: hook, grate

Memory 2 agent chat

Haunted Manor hint: after obtaining/reading the lockbox code note, the next target is to reach/get past/over the heavy iron grate in the parlor, not to focus first on the office painting or hook/chimney.
(Interview note: Human clarified the intended continuation target after the failed run, which changes future strategy and avoids wasting commands.)

Planned features and implementations

1. Alternative to OpenRouter API for better provider flexibility and model variety
2. Usage of already implemented game state management to set up controlled run states for specific model comparisons
3. Advanced statistical tools for comparing models over multiple runs, including table sorting and grouping, diagrams with variance bars and additional logging functions

Contact

For questions, feedback and suggestions, please email me at escape-room-agent@pm.me.